Subashanan Nair

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with 7+ years of experience across machine learning, data engineering and technical education. Builds production data pipelines, dashboards and LLM powered internal tools for Penang Institute and Forward College. Hands-on with Python, scikit-learn, TensorFlow, Streamlit, Docker and GitHub Actions. DataCamp certified AI Engineer for Data Scientists and Kaggle Top 100 competitor.

SKILLS

Languages	Python, SQL, R, JavaScript, TypeScript
ML & AI	scikit-learn, TensorFlow, Keras, XGBoost, LightGBM, CatBoost, Deep Learning, Reinforcement Learning, LLM applications (Ollama, prompt design, RAG, text to SQL)
Data & MLOps	Pandas, NumPy, Streamlit, Plotly, Matplotlib, Seaborn, MLflow, Docker, GitHub Actions, Tableau, Google Drive API
Web & Databases	Next.js, FastAPI, Node.js, REST API design, PostgreSQL, Supabase, Quarto
Tools	Git, GitHub, Jupyter Notebook, Streamlit Cloud

EXPERIENCE

Forward College | *Instructor & Data Scientist* Feb 2019 to Present

- Instruct 3 cohorts per year in Python, machine learning and data analytics, averaging a 4.7/5 learner satisfaction score.
- Built internal programme tools: an LLM assisted project review dashboard, a GitHub based submission and assessment platform, and an exam monitoring app.
- Engineered and deployed ML pipelines and predictive models with Python, scikit-learn and MLflow for client organisations, reducing manual analysis time by 60%.
- Delivered data solutions and dashboards for 3+ organisations, including Penang Institute, enabling data driven policy decisions.

Penang Institute | *Data Scientist (Consultant)* Jun 2024 to Present (2 yrs 4 mos)

- Diagnosed a stale public statistics site and replaced a 34 script R workflow with an automated Python pipeline (GitHub Actions, Docker) that publishes OpenDOSM and data.gov.my data to Google Drive for Tableau.
- Built a weekly ingestion pipeline for key Penang indicators (population, GDP, employment, CPI, household income) backed by a 31 test validation suite.
- Created a pull request gated Streamlit uploader so staff can add new publications without touching code, and an interactive key statistics dashboard (Streamlit, Plotly) deployed on Streamlit Cloud.
- Maintained the Quarto statistics website and delivered a client requested website revamp.

Kuen Lai International Secondary School | *STEAM Consultant* May 2013 to Jan 2019

- Taught Python programming, Arduino and Raspberry Pi electronics, and 3D printing through hands-on, project based STEAM lessons.
- Designed and delivered STEAM programmes aligned with national and international curriculum frameworks.
- Ran workshops for educators on STEAM pedagogy and digital literacy.

PROJECTS

Forward College Project AI Review Tool | *Internal Tool*

Python, Streamlit, GitHub API, Ollama Cloud (gpt-oss:120b), GitHub Actions

- Monitors 9 student capstone repositories and scores each against a 10 section rubric. The LLM reviewer must cite exact source quotes as evidence; fabricated quotes are rejected and the final percentage is computed locally, not taken from the model. Runs hourly on GitHub Actions and never executes student code.

FC Submission Platform | *Internal Tool*

Next.js, TypeScript, Supabase, GitHub OAuth, Playwright

- Students register a public GitHub repo; the platform snapshots the commit at the deadline and runs an automated deliverable checklist. Instructors can override results and grant exceptions with a full audit trail. LLM rubric grading is specified for the next phase.

Exam Guard, Exam Monitoring App | *Internal Tool*
Rust, Tauri, TCP/UDP networking, GitHub Actions

- LAN proctoring tool needing no accounts or internet: a live grid of student screens with problem tiles surfaced first, UDP discovery and class codes. Ships cross platform releases for Windows, macOS and Linux.

Penang Economic Data Assistant | *Prototype*
FastAPI, PostgreSQL, Next.js, Docker, Ollama Cloud

- Natural language assistant over OpenDOSM CPI and unemployment data. Questions are translated to governed SQL with an audit trail, alongside a Next.js data explorer.

Penang Pulse | *Spatial Analysis*
Python, NumPy, SciPy, NetworkX, rasterio, Shapely, pyosmium, OpenStreetMap, WorldPop, DOSM data, Next.js

- Road distance from every populated 100 m cell in Penang to the nearest of 2,437 public clinics, schools and bus stops. 20% of Seberang Perai Selatan residents live over 3 km from a clinic; 74% of all residents live within 1 km of a bus stop.

Forecasting Lab | *Predictive Modelling*
Python, scikit-learn, NumPy, SciPy, PyTorch, Chronos, UCI dataset 608

- Leakage safe 15 minute traffic forecasting benchmark across a 36 sensor network. Gradient boosting cut MAE 23% (95% interval 16 to 30%) versus persistence at 15 minutes; zero-shot Chronos beat it by 26% at 2 hours.

LLM Eval Lab | *LLM Evaluation*
Python, Ollama Cloud, tool calling, PyMuPDF, bootstrap intervals, data.gov.my, DOSM, BNM, MYNIC

- Tested whether cloud LLMs find Malaysia's latest official figure when open data lags: 30 DOSM, BNM and MYNIC statistics, 120 verified questions; with tools, DeepSeek V4.1 Flash gave the latest figure on 91% of out-of-date questions, versus 0% for the open data alone.

Documentation MCP Server
JavaScript, Node.js, Docker, REST API

- Aggregates live documentation from open source libraries (React, Vue, Angular and more) into a searchable REST API and web UI, with daily cron updates and Docker Compose deployment.

EDUCATION

Asia e-University | *Master of Education in Science* — May 2014 to Oct 2016
Universiti Tun Abdul Razak | *Bachelor of Education (Hons)* — May 2009 to Sep 2012

CERTIFICATIONS

DataCamp | *AI Engineer for Data Scientists Associate* — Apr 2026
Certificate ID: AEDS0019212905886
Coursera (DeepLearning.AI & Stanford) | *Supervised ML: Regression and Classification* — 2024
Google for Education | *Google Certified Educator Level 1* — Dec 2021
Kaggle Competition | *Top 100 placement in a global machine learning competition*